3. Data and Methods

3.1 Data Sources

Developing and evaluating the proposed system for job matching and resume tailoring requires real texts of IT candidate profiles and IT job postings. Two open-source datasets from Hugging Face were selected for this study. Both are English parts of the Djinni Recruitment Dataset [1]. It contains anonymized CVs and job vacancies published between 2020 and 2023 on Djinni, a job search platform for the IT sector. Web scraping was not used for data collection.

- **Candidate profiles** (`lang-uk/recruitment-dataset-candidate-profiles-english` [2], MIT license): 210,250 CVs. Each record contains the desired position, the main specialization, years of experience, English level and the full profile text. It serves as the database of candidate profiles.
- **Job descriptions** (`lang-uk/recruitment-dataset-job-descriptions-english` [3], MIT license): 141,897 job postings. Each record contains the position, the company name, the required experience and English level, the month of publication and the full text with responsibilities and requirements. It serves as the database of job offers.

These datasets were chosen because they contain real profiles of IT specialists, which is the target group of the proposed system, and because CVs and jobs come from the same platform and use the same list of specializations. The data has some limitations: it describes mostly the Ukrainian and European IT market, it covers the years before generative AI tools became common, and candidate profiles are short (median 102 words).

3.2 Data Formats and Storage

To keep the project organized and make the results easy to reproduce, a clear folder structure and file naming system were created. Raw data is stored separately from processed data:

- `data/raw/` contains the original files exactly as downloaded from Hugging Face, in Parquet format, together with the original `README.md` files that contains the license. These files are not changed.
- `data/processed/` contains processed data in CSV format. File names have version numbers, like `candidates_it_v1.csv`, so that older versions are kept.
- `data/samples/` contains 50 rows of each processed file, so the data can be viewed without downloading the full datasets.

- `data/docs/` contains a `README.md` file and the codebook (`codebook.md`) that describes the datasets, their columns and data types.
- `notebooks/` contains Jupyter notebooks with all processing steps. The data selection is done in `notebooks/01_data_selection.ipynb`.

The raw Parquet files can be downloaded from the links given in Section 3.1, and all processed files are created from them by running the notebooks. All random steps use a fixed seed (`random_state = 5267`), so the same files are produced every time.

Tables 3.1 and 3.2 give a short codebook of both datasets. The full version with value ranges and missing values is stored in `data/docs/codebook.md`. The technical column `__index_level_0__`, left from the original export, has no meaning and is removed.

Table 3.1: Codebook of the candidate profiles dataset

Column	**Type**	**Meaning**
`id`	string (UUID)	Unique record identifier
`Position`	text	Desired job title written by the candidate
`Primary Keyword`	categorical	Main specialization selected on the platform (e.g. Java, QA, DevOps)
`Experience Years`	numeric	Years of commercial experience, from 0 to 11
`English Level`	ordinal	Self-reported English level, from `no_english` to `fluent`
`Highlights`	text	Main achievements (optional)
`Moreinfo`	text	Main “about me” section: experience, projects, skills
`Looking For`	text	Expected job or team (optional)
`CV`	text	Full profile text, a concatenation of `Highlights`, `Moreinfo` and `Looking For`, the main field used for matching
`CV_lang`	categorical	Language of the text, always `en`

Table 3.2: Codebook of the job descriptions dataset

Column	Type	Meaning
id	string (UUID)	Unique record identifier
Position	text	Job title of the posting
Company Name	text	Name of the hiring company
Primary Keyword	categorical	Main specialization of the posting, same values as for candidates
Exp Years	ordinal	Required experience: no_exp, 1y, 2y, 3y, 5y
English Level	ordinal	Required English level, from basic to fluent
Published	date (ISO 8601)	Month of publication, from January 2020 to September 2023
Long Description	text	Full job posting: company, responsibilities, requirements, the main field used for matching
Long Description_lang	categorical	Language of the text, always en

3.3 Preprocessing and Cleaning

Since the system uses Natural Language Processing (NLP) and Sentence-BERT embeddings, the raw data needed to be cleaned before use. The processing was done in Python using the pandas library.

First, the data was filtered to focus on the IT sector. The full dataset covers all professions present on the platform, including design, marketing, sales and human resources. The selection is based on the Primary Keyword field. Out of 41 specializations, 23 IT specializations were kept and grouped into eight role families, which are stored in a new column role_family (Table 3.3). Non-technical roles (for example Design, Marketing, HR, Sales, Project Manager) and unclear groups (Other, Lead) were removed.

Table 3.3: Role families and number of records after selection

Role family	Primary Keyword values	Candidates	Jobs
Backend	Java, .NET, Python, PHP, Node.js, Golang, Ruby, Scala, Rust, C++	46,510	8,826
Frontend / JavaScript	JavaScript	33,980	3,831
QA	QA, QA Automation	30,845	2,910
Mobile	iOS, Android, Flutter	7,405	1,091
Data / ML	Data Science, Data Engineer, Data Analyst, SQL	6,764	1,268
DevOps	DevOps	4,120	1,671
GameDev	Unity	1,815	247
Security	Security	781	156
Total		**132,220**	**20,000**

Second, the number of job postings was reduced. After the selection, 95,474 IT job postings remained. Encoding all of them with Sentence-BERT on a laptop would take several hours, while 20,000 postings are enough to compare the search methods. Therefore, a random sample of 20,000 job postings was taken (`random_state = 5267`). The sample keeps the role proportions of all IT job postings: the difference is below 0.5 percentage points for every role family. Table 3.4 shows the number of records after each step.

Table 3.4: Number of records after each selection step

Step	Candidates	Jobs
Raw English dataset	210,250	141,897
IT roles only	132,220	95,474
Random sample of job postings	132,220	20,000

The selected data is saved as `candidates_it_v1.csv` and `jobs_it_v1.csv`. The texts themselves are not changed at this stage. A first look at the files showed some problems: duplicated and very short profiles, broken characters, a few phone numbers and links to personal profiles, and profiles with a wrong specialization (for example, some 2D animators selected `Unity`).

The text is prepared differently for each part of the system: BM25 uses lowercase text without punctuation because it searches for exact words, while Sentence-BERT and the CV personalization module keep capital letters and punctuation because they need full sentences to understand the meaning and generate a correct CV. Both the cleaning and this preparation are described in Chapter 4.

3.4 Annotations and Labels

The system uses cosine similarity to compare CVs and job descriptions. Although it can do this without labeled data, examples with known results are needed to check how well the system works.

The Djinni dataset has no information about which candidate fits which job, so the labels are created automatically from the specialization chosen on the platform (`Primary Keyword`). A job is relevant if it has the same specialization as the CV and partly relevant if it belongs to the same role family. Since Sentence-BERT is pre-trained and BM25 needs no training, no training set is used. Instead, 100 CVs from the cleaned data are used as a development set and 500 CVs as a test set for Precision@10 and nDCG@10, sampled evenly over the eight role families (`random_state = 5267`).

The CV adaptation module also needs to be tested to make sure the generated CVs are suitable for professional use. For 30 test CVs, the match with the target job is evaluated both before and after adaptation, and each adapted CV is scored by the author from 1 to 5 on three criteria:

- **Factual Accuracy:** Ensuring the model does not hallucinate or add fake skills.
- **Alignment:** Highlighting the specific background that matches the target vacancy.
- **Readability:** Checking for natural phrasing and proper grammar.

3.5 Ethical and Legal Considerations

The AI tools used for recruiting need to stick to all privacy laws and ethical standards. Both datasets that were used in this study are publicly available under the MIT license, which allows using them for research. The license is located near datasets as `README.md` files.

The CVs and job postings were anonymized by the dataset authors before publication [1]. A check found no e-mail addresses, but a few phone numbers and links to personal profiles (LinkedIn, GitHub, Telegram) were left. They are deleted during cleaning in Chapter 4.

The system will help users find suitable job offers and improve their resumes. However, the tool will not be used for making hiring decisions. Considering that LLMs can show some bias, the final decision will be up to users and recruiters. The users will also get the chance to check the offered job opportunities and resume before using them.

3.6 Presentation of Results

The results will be presented in tables and figures. Tables will show numerical results, such as Precision@10 and nDCG@10 of BM25 and Sentence-BERT, similarity scores and system response time. The work of the language model will be shown with examples of CVs before and after changes. Figures will be used to show how the system works, how the data is distributed, and how different scoring systems affect the results. Each table and figure will have a number and a short title. They will also be mentioned in the text to help readers understand the results.

References

1. N. Drushchak and M. Romanyshyn, *"Introducing the Djinni Recruitment Dataset: A Corpus of Anonymized CVs and Job Postings,"* in Proceedings of the Third Ukrainian Natural Language Processing Workshop (UNLP) @ LREC-COLING 2024, Torino, Italy, pp. 8–13, 2024. URL: https://aclanthology.org/2024.unlp-1.2

2. lang-uk, *"Djinni Dataset (English CVs part),"* Hugging Face dataset, MIT license. URL: https://huggingface.co/datasets/lang-uk/recruitment-dataset-candidate-profiles-english

3. lang-uk, *"Djinni Dataset (English Job Descriptions part),"* Hugging Face dataset, MIT license. URL: https://huggingface.co/datasets/lang-uk/recruitment-dataset-job-descriptions-english